Swarm basic node commands

Every command the Swarm app sends to a basic node, and every line it understands back.

> A **basic node** is any board that advertises the node service below and speaks plain text lines, but doesn't run Swarm node firmware. It scans whole bands on command and streams what it sees live. It gets no channel slot and keeps nothing while out of range.

1. Connection

What	Value
Service (advertised)	`4fafc201-1fb5-459e-8fcc-c5c9c331914b`
Command (app writes here)	`beb5483e-36e1-4688-b7f5-ea07361b26a8`
Response (node notifies here)	`beb5483e-36e1-4688-b7f5-ea07361b26a9` The app turns notifications on (CCCD `0x0001`). Send nothing before that.
MTU	The app asks for 247. It carries on at the default MTU if the node doesn't answer.

Every message is one text line: `TYPE:SUB:PAYLOAD` ending in `\n`. The colon after SUB stays even when the payload is empty (`CMD:stopscan:`). A notification may hold several lines, and one line may span several notifications, so split on `\n`. A trailing `\r` is ignored.

2. Commands the app sends (app to node)

Line	What it means	When the app sends it
`CMD:hello:v=6`	Asks whether the board runs Swarm firmware.	Right after connecting. A basic node should answer `RSP:hello:ERROR` (or nothing: after 4 seconds the app treats it as basic).
`CMD:wardrive:`	Scan 2.4 GHz Wi-Fi.	Run started, plan gives this node 2.4 GHz only.
`CMD:wardrive5g:`	Scan 5 GHz Wi-Fi.	Run started, plan gives this node 5 GHz only.
`CMD:btwardrive:`	Scan Bluetooth (BLE).	Run started in BLE-only mode, or the node has no Wi-Fi work.
`CMD:wardriveall:`	Scan everything (2.4 + 5 GHz Wi-Fi and BLE).	Run started with Wi-Fi and BLE, or with both Wi-Fi bands.
`CMD:stopscan:`	Stop any scan.	Run stopped. Also sent before switching a node from one scan command to another.
`TIME:SYNC:2026-10-16T02:30:00Z`	The current time, UTC, ISO-8601.	Every time the node sends `TIME:REQUEST:`.

3. Lines the app understands (node to app)

Line	Meaning
`RSP:<cmd>:OK`	Command accepted (not finished).
`RSP:<cmd>:ERROR[:reason]`	Command refused. If a scan command is refused, the app waits 30 seconds before trying again.
`STATUS:<code>:<text>`	Node state. The app uses 1 ready, 2 busy (scanning), 6 scan complete. Other codes (0 booting, 3 ok, 4 warning, 5 error) are accepted and ignored.
`TIME:REQUEST:`	Node wants the time. The app answers with `TIME:SYNC:`.
`DATA:AP:SSID,BSSID,Channel,RSSI,Security`	One Wi-Fi network. Example: `DATA:AP:MyWiFi,AA:BB:CC:DD:EE:FF,6,-58,[WPA2_PSK]`
`DATA:BT:Name,MAC,RSSI,Type`	One Bluetooth device. Extra fields after Type are allowed and ignored. Example: `DATA:BT:Tag,11:22:33:44:55:66,-71,BLE`
`DATA:NODEAP:...` `DATA:NODEBT:...`	Same as AP / BT, with the relaying board's MAC added at the end. Treated exactly like AP / BT.
`DATA:MESH:Network,NodeId,Name,RSSI[,Role][,pk=..][,snr=..]`	A mesh radio node the board heard. Network is `meshcore` / `meshtastic` (or `MC` / `MT`).

Rules for data lines:

- MACs are `AA:BB:CC:DD:EE:FF` form; lines with a bad MAC are dropped.
- Channel must be 1 to 196 and RSSI -128 to 127, or the line is dropped.
- Replace `,`, `\r`, `\n` in names and SSIDs with `_`.
- No GPS or time is needed in the line: the phone stamps each sighting with its own time and location when it arrives.

Security values for DATA:AP

Send	Logged as
`[OPEN]` or empty	`[ESS]`
`[WEP]`	`[WEP][ESS]`
`[WPA_PSK]`	`[WPA-PSK-TKIP][ESS]`
`[WPA2_PSK]`	`[WPA2-PSK-CCMP][ESS]`
`[WPA_WPA2_PSK]`	`[WPA-PSK-TKIP][WPA2-PSK-CCMP][ESS]`
`[WPA2_ENTERPRISE]`	`[WPA2-EAP-CCMP][ESS]`
`[WPA3_PSK]`	`[WPA3-SAE-CCMP][ESS]`
`[WPA2_WPA3_PSK]`	`[WPA2-PSK-CCMP][WPA3-SAE-CCMP][ESS]`
`[WPA3_ENTERPRISE]`	`[WPA3-EAP-CCMP][ESS]`
`[OWE]`	`[OWE][ESS]`

Brackets are optional and case doesn't matter. Anything else is logged as `[VALUE][ESS]`.

4. How the app drives a basic node

- Connect, turn on notifications, send `CMD:hello:v=6`. No Swarm heartbeat within 4 seconds, or `RSP:hello:ERROR`, makes it a basic node with its own node number.
- When a run starts, the app sends the one scan command that covers the node's part of the plan.
- While the run is going, if the node reports `STATUS:1` or `STATUS:6` (idle), the app sends the scan command again, at most every 10 seconds.
- If the plan changes the node's band, the app sends `CMD:stopscan:` then the new scan command.
- When the run stops, the app sends `CMD:stopscan:`.
- After a dropped link, the app reconnects and sends the scan command again; the node doesn't need to remember anything.
- Sightings are only logged while a run is going, and only for the sensors picked (Wi-Fi and/or BLE). Mesh nodes are always kept.